

Ten Papers: Physical Formulas Are Not Discovered, but Defined

Paper One — Following Einstein, Introduction, Addition, and Subtraction

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Abstract

Following the attempts of Einstein and other predecessors to establish the foundation of a unified physical theory, this series of studies puts forward a conjecture: physical formulas are not merely discovered through observation, but defined following logical rules. This viewpoint inherits Einstein's conventionalism in physical theories and is grounded in academic philosophy, rather than pseudoscience.

This is the first paper in the ten-paper series. The whole research aims to construct a standardized framework based on original insights, revealing the meta-rules governing the construction of physical formulas and reducing blind empirical fitting and heuristic speculation. Due to the complexity of axiomatic deduction, each type of arithmetic operation is discussed in an independent paper.

This paper focuses on independent physical subsystems and proves the logical necessity of addition and subtraction when aggregating independent physical components. Only the relationships among scalar and vector independent components are analyzed here. The ultimate conclusion of the entire series will be presented as a conjecture in the final paper, which does not affect the logical rigor and validity of the arguments in this article. This paper only discusses addition and subtraction. Subsequent papers will extend the research to multiplication, division, exponents and other arithmetic operations.

1. Introduction

Physical equations are mathematical mappings of the interactions between matter, motion and spacetime, instead of arbitrary numerical fitting (Feynman, Leighton, & Sands, 1964). Traditional physical modeling largely relies on trial-and-error observation and personal intuition to determine equation structures, which raises the learning threshold for students and young researchers (Knight, 2022). To establish a repeatable, rule-driven process for

generating physical formulas, each fundamental arithmetic operation needs to be verified as an irreducible primitive rule.

Physicists and philosophers of science including Einstein and Poincaré have long argued that the fundamental concepts and principles of physics are not completely determined by experience, and theoretical systems contain prominent conventional elements. Re-examining Kant's theory of the a priori, Einstein interpreted "the a priori" as "the conventional". He repeatedly stated that the logical framework and internal structure of physical theories are free creations of the human mind. The value of such theories lies in their consistency with empirical facts and internal logical unity (Einstein, 1949).

At an earlier stage, Poincaré proposed a classic view on the foundations of geometry: geometric axioms are neither synthetic a priori judgments nor pure empirical facts, but human conventions. Subject to empirical constraints, people choose geometric systems based on simplicity and practicality (Poincaré, 1902/1952). Einstein further extended this methodology from geometry to the whole field of physics.

Based on the above theoretical background, this series explores the underlying rules of formula construction starting from the expression of physical formulas. This paper addresses the most fundamental question: Why do we use addition and subtraction, rather than multiplication and division, to aggregate independent physical components?

This paper first formalizes the definition of independent physical components, deduces the intrinsic logic of using addition for components acting in the same direction and subtraction for opposing components, and verifies the rules with classical physical scenarios and conclusions from mainstream textbooks.

2. Formal Definition of Independent Physical Subsystems

Suppose a composite system is divided into measurable scalar physical quantities $Q_1, Q_2, \dots, Q_n$. A pair of components (Q_i, Q_j) is defined as independent if all three criteria below are satisfied:

1. Invariance of magnitude: Adding or removing component Q_j will not change the intrinsic magnitude of Q_i (Halliday, Resnick, & Walker, 2021).

2. Zero interaction: No transfer of mass, energy, momentum, charge or field energy exists between components (Serway & Jewett, 2020).
3. Full contribution: Each independent component contributes its complete magnitude to the total observable of the system, without amplification, attenuation or occupation of the contribution of other components.

Addition is adopted to aggregate independent components acting in the same direction, while subtraction is used for independent components with opposite directions and offset effects.

For the independent system defined in this paper, multiplication and division are logically invalid. If we apply multiplication $T = Q_1 \cdot Q_2$, the total magnitude will become zero when $Q_2 = 0$. This violates the invariance criterion that the contribution of remaining components shall stay unchanged after one component is removed. Similarly, division cannot meet the constraints of independent systems. Therefore, addition and its signed extension (subtraction) are the only logically consistent operations for independent components.

Note: This conclusion is only applicable to fully independent subsystems defined in this paper. Systems with coupled components will be discussed exclusively in the second paper of this series.

3. Axiomatic Deduction of Addition Rules

3.1 Two-component parallel independent system

Consider two fully independent physical components Q_1 and Q_2 acting in the same direction.

1. State 1 (Only Q_1 exists): Total output $T_1 = Q_1$
2. State 2 (Only Q_2 exists): Total output $T_2 = Q_2$
3. State 3 (Both components coexist): In accordance with the independence criteria, the magnitudes of Q_1 and Q_2 remain unchanged, and both deliver full contributions. The only logically consistent aggregation is summation:

$$T = Q_1 + Q_2$$

3.2 Generalization to n parallel independent components via mathematical induction

Base case: The case for $n = 2$ has been proven valid.

Inductive hypothesis: For k independent components acting in the same direction, the total magnitude of the system satisfies:

$$T_k = \sum_{i=1}^k Q_i$$

When a new independent component Q_{k+1} is added:

$$T_{k+1} = T_k + Q_{k+1} = \sum_{i=1}^{k+1} Q_i$$

It is proved that for any integer $n \geq 2$, the aggregation of parallel independent components follows the summation rule. No additional scaling factor is required, and the inherent proportional coefficient equals 1 (Young & Freedman, 2020).

Scope of application: Mass superposition in classical mechanics, total current in parallel DC circuits, energy summation of non-interacting systems, and composition of forces in the same direction.

3.3 Supplement: Orthogonal vector independent components

For independent components expressed as spatial orthogonal vectors (e.g., components along the x -axis and y -axis in a plane rectangular coordinate system), components on each axis still follow the rules of addition and subtraction. The resultant vector is finally obtained through geometric vector composition. The operation rules for individual axes are identical to those for scalar quantities, and orthogonality only changes the final form of results without altering the fundamental logic of addition and subtraction.

4. Axiomatic Deduction of Subtraction Rules

Mathematically, subtraction is the signed extension of addition for components with opposite polarities. From a physical perspective, positive and negative signs correspond to specific physical meanings such as acting directions, energy budget and charge properties, rather than pure mathematical symbols.

4.1 Two-component opposing independent system

Let Q_+ denote the component with positive contribution and Q_- denote the offset component with negative contribution. The two components still satisfy all independence criteria: neither will change the magnitude of the other, and only their overall effects counteract each other.

1. Isolated positive contribution: $T_+ = Q_+$
2. Isolated offset contribution: $T_- = Q_-$
3. Net output with both components coexisting: The net magnitude equals the positive contribution minus the offset part:

$$T_{\text{net}} = Q_+ - Q_-$$

4.2 Generalization for net offset of multiple components

Divide all independent components into a positive group $\{Q_{+1}, Q_{+2}, \dots\}$ and an offset group $\{Q_{-1}, Q_{-2}, \dots\}$. The net total magnitude of the system is written as:

$$T_{\text{net}} = \sum Q_+ - \sum Q_-$$

Essentially, subtraction is not an independent primitive operation, but an extension of the aggregation rule for independent components under opposing polarities.

Scope of application: Net collinear forces, net charge of a system, energy inflow and loss, and counteracting currents.

5. Cross-disciplinary Physical Verification and Benchmark Cases

5.1 Mechanics: Collinear independent forces (Addition and Subtraction)

A wooden block is acted upon by a rightward pulling force F_1 and a leftward kinetic friction force F_f . The two forces are independent, and their magnitudes will not be affected by each other (Halliday et al., 2021, p. 89).

The net force of the system is $F_{\text{net}} = F_1 - F_f$. If two pulling forces act in the same direction, the total force is $F_{\text{total}} = F_1 + F_2$. Experimental measurements from force sensors verify the validity of addition and subtraction for linear force superposition (Knight, 2022, p. 126).

5.2 Conservation of mass (Only addition is applicable)

Consider two solid objects with no interaction or material exchange, with masses m_1 and m_2 respectively. Negative mass does not exist in classical physics, and mass is an inherently non-negative physical quantity. Thus the total mass is calculated only by addition: $M = m_1 + m_2$ (Serway & Jewett, 2020, p. 201).

5.3 Thermodynamics: Decomposition of internal energy

The internal energy of an ideal gas can be decomposed into mutually independent translational energy U_t , rotational energy U_r and vibrational energy U_v (Feynman et al., 1964, Vol. 1, p. 412). The total internal energy is $U = U_t + U_r + U_v$. If heat loss Q_{loss} is regarded as an independent offset term, the net retained energy of the system is $U_{\text{net}} = U_{\text{in}} - Q_{\text{loss}}$.

5.4 DC circuits: Current in parallel branches and net charge

In a DC parallel circuit, currents I_1 and I_2 in different branches are independent of each other. The total supply current is $I = I_1 + I_2$. If leakage current I_{leak} causes energy loss, the net current at the load end is $I_{\text{net}} = I_{\text{supply}} - I_{\text{leak}}$ (Young & Freedman, 2020, p. 603).

6. Summary of Application Conditions for Addition and Subtraction

Condition	Operation	Examples
Independent components acting in the same direction	Addition	Forces in the same direction, current in parallel circuits, superposition of non-interacting masses
Independent components acting in opposite directions	Subtraction	Opposing collinear forces, net charge, energy budget calculation
Independent components in	Addition and subtraction on each axis, followed by	Composition of two-dimensional forces

Condition	Operation	Examples
orthogonal directions	geometric vector composition	
Components with mutual coupling	Addition and subtraction are invalid (discussed in subsequent papers)	Systems requiring multiplication and other operations

Special Cases and Boundary Limitations

1. Restriction of operations: In classical physics, subtraction is only applied to physical quantities that can take both positive and negative values, such as force, current and energy variation. It is not suitable for inherently non-negative quantities including mass and time interval.
2. Deviation in real systems: Fully independent ideal systems barely exist in nature. Weak residual coupling will generate high-order perturbation terms, but will not change the dominant addition-subtraction structure of the system. Relevant analysis on high-order corrections will be presented in follow-up papers.

Scope of the framework: The model established in this paper only applies to addition and subtraction for fully independent subsystems. Coupled systems, relativistic systems and quantum systems involve modified or extended operation rules, which are beyond the scope of this paper.

7. Boundary Statement of the Ideal Model

The deduction of addition and subtraction in this paper is based on the ideal model of fully independent systems, which describes the dominant structure of physical formulas. Weak coupling commonly found in real physical systems leads to minor deviations as high-order perturbations, which do not invalidate the core rules of addition and subtraction (Feynman et al., 1964, Vol. 1, p. 450). Corrections for residual coupling will be discussed in subsequent papers of this series.

8. Conclusions of Paper 1

1. Addition is applied to superposition of parallel, co-directional and mutually independent physical quantities.
2. Subtraction is applied to counteraction of parallel, opposing and mutually independent physical quantities.
3. For the independent component system defined in this paper, multiplication and division violate the independence criteria. Therefore, addition and subtraction are the only logically consistent operations.
4. A large number of fundamental formulas in classical physics follow the above rules: conservation of mass $M = m_1 + m_2$, net opposing force $F_{\text{net}} = F_1 - F_2$, total current in parallel circuits $I = I_1 + I_2$, and superposition of multi-mode internal energy $U = \sum U_i$.
5. The application of subtraction is limited by the properties of physical quantities. Weak coupling in real systems only produces high-order corrections and does not affect the main structure of formulas.
6. Addition and subtraction form the first set of primitive operations in the operational system of physical formulas. The rules have been verified through axiomatic deduction, mathematical induction and cross-disciplinary case analysis.

9. Preview of Paper 2

The second paper will focus on coupled physical systems and answer the core question: When physical quantities are no longer independent and interact with each other, how does the multiplication rule in physical formulas come into being? Subsequent papers will prove that multiplication becomes the only logically consistent operation when physical quantities meet specific coupling criteria.

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